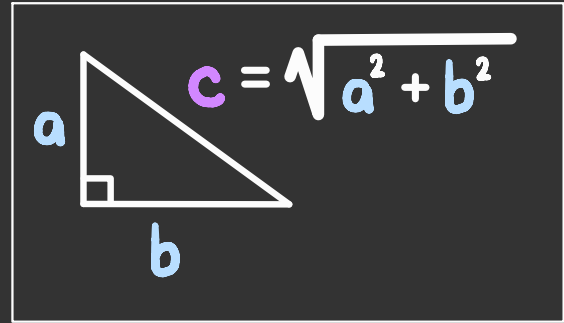


inner
functions

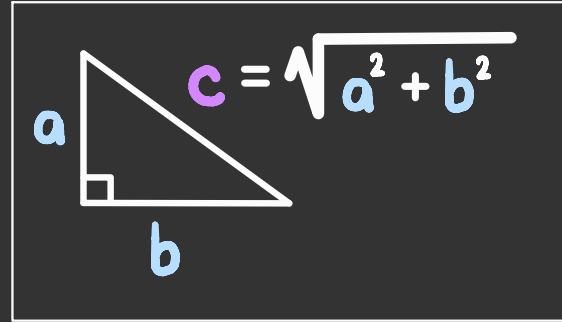
CSCI
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one way to compute the
pythagorean theorem



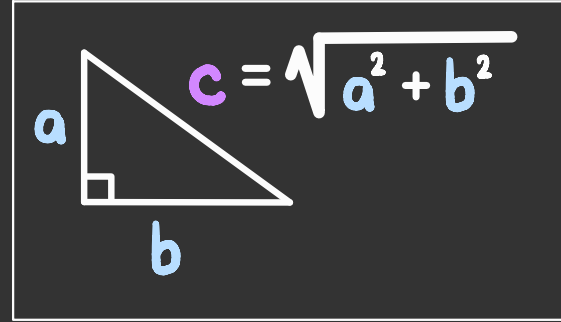
```
def pythagorean(a, b):  
    square = lambda x: pow(x, 2)  
    square_root = lambda x: pow(x, 0.5)  
    return square_root(square(a) + square(b))
```

but we know another way
to **define** functions



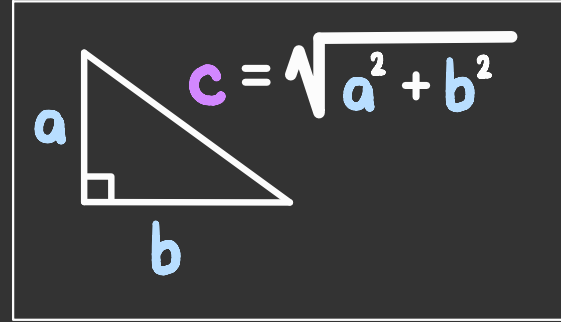
```
def pythagorean(a, b):  
    square = lambda x: pow(x, 2)  
    square_root = lambda x: pow(x, 0.5)  
    return square_root(square(a) + square(b))
```

this also works



```
def pythagorean(a, b):  
    def square(x):  
        return pow(x, 2)  
    def square_root(x):  
        return pow(x, 0.5)  
    return square_root(square(a) + square(b))
```

these are called
inner functions



```
def pythagorean(a, b):  
    | def square(x):  
    |     return pow(x, 2)  
    |  
    | def square_root(x):  
    |     return pow(x, 0.5)  
    |  
    | return square_root(square(a) + square(b))
```

←

←

```
oscar = 333444333
```

```
def dorm():
```

```
    zoe = 555999555
```

```
        def room1():
```

```
            elmo = 123456789
```

```
            print(elmo)
```

```
            print(zoe)
```

```
            print(oscar)
```

```
        def room2():
```

```
            oscar = 567891234
```

```
            ernie = 112233445
```

```
            print(ernie)
```

```
            print(zoe)
```

```
            print(oscar)
```

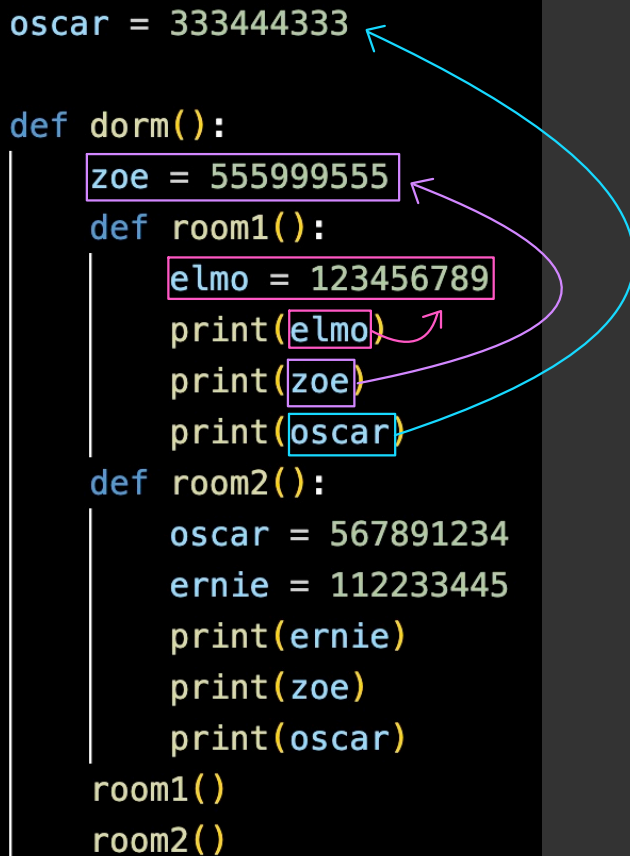
```
    room1()
```

```
    room2()
```

how does variable
scope work for
inner functions?

```
oscar = 333444333

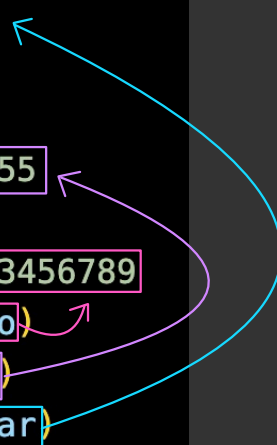
def dorm():
    zoe = 555999555
    def room1():
        elmo = 123456789
        print(elmo)
        print(zoe)
        print(oscar)
    def room2():
        oscar = 567891234
        ernie = 112233445
        print(ernie)
        print(zoe)
        print(oscar)
    room1()
    room2()
```



From **most** local
to **least** local

```
oscar = 333444333

def dorm():
    zoe = 555999555
    def room1():
        elmo = 123456789
        print(elmo)
        print(zoe)
        print(oscar)
    def room2():
        oscar = 567891234
        ernie = 112233445
        print(ernie)
        print(zoe)
        print(oscar)
    room1()
    room2()
```



In [2]: dorm()

123456789

555999555

333444333

112233445

555999555

567891234

```
oscar = 333444333
```

```
def dorm():
```

```
    zoe = 555999555
```

```
    def room1():
```

```
        elmo = 123456789
```

```
        print(elmo)
```

```
        print(zoe)
```

```
        print(oscar)
```

```
    def room2():
```

```
        oscar = 567891234
```

```
        ernie = 112233445
```

```
        print(ernie)
```

```
        print(zoe)
```

```
        print(oscar)
```

```
    room1()
```

```
    room2()
```

In [2]: dorm()

123456789

555999555

333444333

112233445

555999555

567891234